AI101 MINI-PROJECT · GROUP 68

Machine Learning-Based Prediction of Airfoil Self-Noise

Akshaj Mishra
PPT Lead

Arghya Mehta
Execution Lead

Himanshu Prajapati
Documentation Lead

Swarnum Thappa
Video Lead

GitHub repository link: https://github.com/ArghyaM-codes/Airfoil-self-noise-prediction

Problem Statement and Dataset

What are we predicting?

Airfoils produce noise when air flows around them. We use machine learning to predict this sound pressure level from operating conditions.

Why does it matter?

Predicting airfoil noise can help engineers understand noise patterns and design quieter aerodynamic systems, such as aircraft components.

Dataset Source

NASA Airfoil Self-Noise Dataset
UCI Machine Learning Repository

Size

1,503
Rows (observations)

6
Columns (5 inputs + 1 target)

Columns in the dataset

- Frequency
- Angle of attack
- Chord length
- Free-stream velocity
- Suction-side displacement thickness

Target column: Sound Pressure Level (dB)

Our Approach – Linear Regression

NASA Airfoil Dataset
1,503 observations

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Data Exploration (EDA)
Statistics, distributions, correlations

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Train-Test Split
80% training · 20% testing

↓

Linear Regression Model
Training on input features

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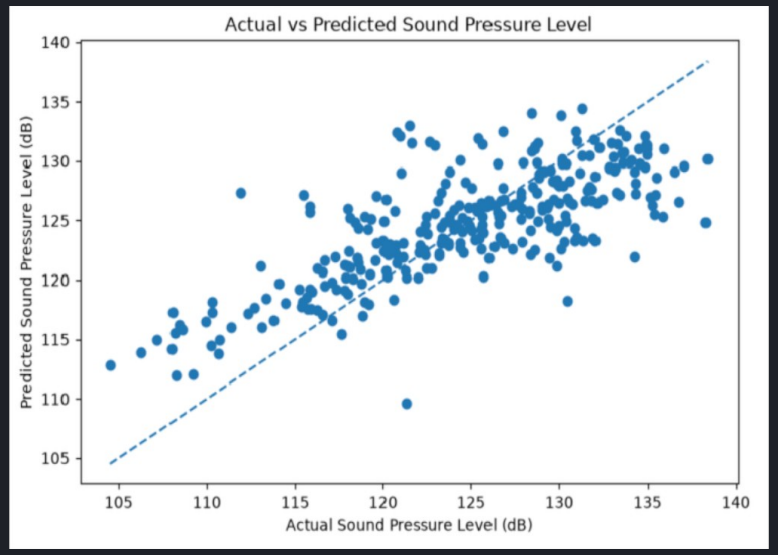
Predicted Sound Pressure Level
Evaluation using MAE, RMSE and R^2

Libraries / tools used: NumPy · Pandas · Matplotlib · Seaborn · Scikit-learn · Jupyter

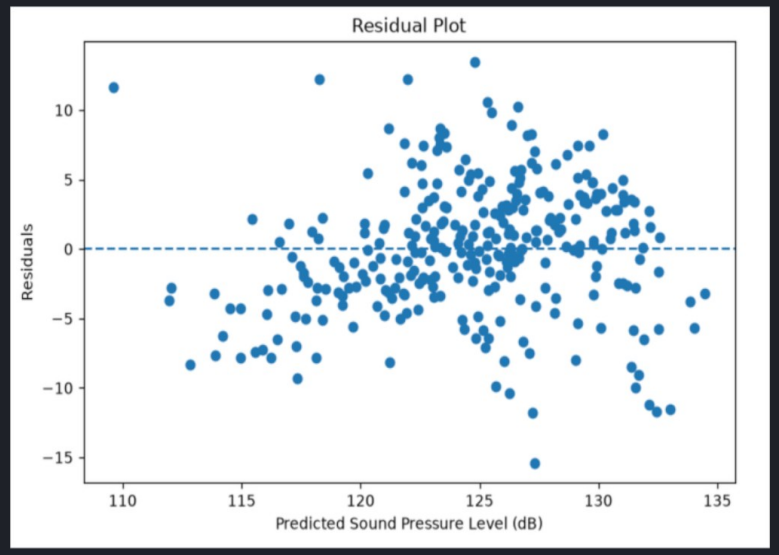
Model Results and Evaluation

MAE	MSE	RMSE	R^2 Score
3.6724 dB	22.1286 dB^2	4.7041 dB	0.5583

Actual vs Predicted



Residual Plot



Demo: Notebook Output

```
184    120.233
1000   137.047
746    134.556
Name: Sound_pressure_level, dtype: float64

Predicted values:
[124.16191238 119.71754202 120.29668459 129.615236   129.75593826]
```

```
In [15]:
mae = mean_absolute_error(y_test, y_pred)

mse = mean_squared_error(y_test, y_pred)

rmse = np.sqrt(mse)

r2 = r2_score(y_test, y_pred)

print("Model Evaluation Results")
```

Other EDA graphs: distributions, correlation heatmap, feature correlations and feature relationships.

Linear Regression captures some of the pattern, but prediction errors remain.

Project Wrap-Up

1. What Worked / What Didn't

What Worked:

- Successful EDA and feature insights.
- Built and evaluated a Linear Regression model.

What Didn't Work:

- Moderate model performance ($R^2 \approx 0.558$).
- Linear Regression missed complex patterns.

2. Who Did What

Akshaj Mishra

PPT Lead

Exploratory Data Analysis (EDA), visualizations, feature correlation, final PPT.

Arghya Mehta

Execution Lead

Python & ML implementation, preprocessing, model evaluation, repository management.

Himanshu Prajapati

Documentation Lead

Project leadership, README, repository structure, utility code (src/utils.py).

Swarnum Thappa

Video Lead

Video presentation, workflow explanation, code testing and debugging.

3. AI-Use Disclosure

Akshaj Mishra

ChatGPT: Coding & EDA guidance, debugging, results interpretation, PPT preparation.

Arghya Mehta

ChatGPT: Python programming, debugging, ML concepts and implementation.

Himanshu Prajapati

Gemini: Repository architecture, README, helper functions, rubric compliance.

Swarnum Thappa

ChatGPT: Understanding workflow, coding, debugging and video planning.